



MOKO SMART



User Guide

MOKO TAG APP

Version 1.2

Revision History

Version	Data	Notes	Contributor(s)
V1.0	Aug 30, 2022	Initial version	Daniel
V1.1	Sept 27, 2022	1. Remove the description of Coded PHY 125Kbps in advertisement.	Daniel
V1.2	May 24, 2023	1. Add " Static Heartbeat " function configuration page under Setting -> Sensor configurations -> 3-axis accelerometer page. 2. Add " Remote reminder " function under Setting page, which is used to remotely control LED on/off of the beacon device. 3. Add " Reset Battery " function under Setting page, which allows users to reset the device's statistical data and restart calculate the battery level after replacing the battery (This function only applies to devices that use ER14505 lithium-ion batteries)	Leo

About this User Guide

This User Guide was designed to help users to know the overview of **MOKO TAG** series products and set up **MOKO TAG** through “**MOKO TAG**” APP. Through this document, users will initial to know the product’s functional framework, the setup process of Beacon advertisement, connection, configuration, DFU update, etc.

This guide will not cover the sales administration and the ordering process. Some additional technical guides will be needed if further explanation is required.

• Applicable version description:

Android APP Version: “MOKO TAG” V1.0.0 or above

iOS APP Version: “MOKO TAG” V1.0.0 or above

Firmware Version: BXP-T_V1.0.0 or above

• Applicable products:

Product model name	Product order name	ID picture
M4 Lite Tag	M4-TSLAA	
M3 Mini	-	

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1. Overview

This User Guide is mainly applicable for MOKO standard **“MOKO TAG”** APP and MOKO Tag with **“BXP-T”** serial firmware, and mainly contained below parts:

- [Content parsing of Beacon advertisement](#)
- [“MOKO TAG” APP user guidance](#)
- [FAQ](#)

For more hardware technical information or assembly/disassembly instructions, please ask our sales team directly for separated document – **“Product Specification”**.

2. Content parsing of Beacon advertisement

Before introducing the APP user manual, we will guide you to fully understand the Beacon advertisement format and content parsing rules. In terms of the MOKO “**BXP-T**” series firmware, which is especially design for asset tracking, it contains several kinds of advertisement frame type which include standard format and customized format, that is:

- *Google Eddystone™ - UID*
- *Google Eddystone™ - URL*
- *Google Eddystone™ - TLM (Unencrypted)*
- *Customized advertisement frame – “iBeacon”*
- *Customized advertisement frame - “Tag info”*

To improve the readability of the screen formats, this document presents:

- The fixed Bluetooth fields highlighted in **blue**.
- The variable Bluetooth fields highlighted in **orange**.
- User-defined/configurable fields highlighted in **green**.
- Reserved fields highlighted in **red**.

The screenshots have been taken on the “**nRF Connect**” application, developed by Nordic Semiconductors.

2.1 Google Eddystone – UID

In terms of UID advertisement frame type, this is exactly complied with Google Eddystone regulations, and you can refer to below link for more information: <https://github.com/google/eddystone/tree/master/eddystone-uid>

Below table described the detailed content structure and frame example of **UID** advertisement:

Eddystone - UID																																																																																	
Frame bytes	Frame format		Frame example																																																																														
	01	Length: 0x02	Received frame : Eddystone - UID <table><tr><td>Namespace ID</td><td>0x0102030405060708090A</td></tr><tr><td>Instance ID</td><td>0x102030405060</td></tr><tr><td>RSSI@0m</td><td>0x00</td></tr></table> Raw data: 0x0201060303AAFE1716AAFE00000102030405060708090A102030405060000 Details: <table><tr><th>LEN.</th><th>TYPE</th><th>VALUE</th></tr><tr><td>2</td><td>0x01</td><td>0x06</td></tr><tr><td>3</td><td>0x03</td><td>0xAAFE</td></tr><tr><td>23</td><td>0x16</td><td>0xAAFE00000102030405060708090A1020304050600000</td></tr></table> LEN. - length of EIR packet (Type + Data) in bytes, TYPE - the data type as in https://www.bluetooth.org/en-us/specification/assigned-numbers/generic-access-profile OK Eddystone - UID field <tr><td>02</td><td>Type: 0x01</td></tr> <tr><td>03</td><td>Data: 0x06</td></tr> <tr><td>04</td><td>Length: 0x03</td></tr> <tr><td>05</td><td>Type: 0x03</td></tr> <tr><td>06</td><td>Eddystone_UUID_LSB: 0xAA</td></tr> <tr><td>07</td><td>Eddystone_UUID_MSB: 0xFE</td></tr> <tr><td>08</td><td>Length: 0x17</td></tr> <tr><td>09</td><td>Type: 0x16</td></tr> <tr><td>10</td><td>Eddystone_UUID_LSB: 0xAA</td></tr> <tr><td>11</td><td>Eddystone_UUID_MSB: 0xFE</td></tr> <tr><td>12</td><td>Frame type UUID: 0x00</td></tr> <tr><td>13</td><td>RSSI@0m: 0x00</td></tr> <tr><td>14</td><td>Namespace ID[0]</td></tr> <tr><td>15</td><td>Namespace ID[1]</td></tr> <tr><td>16</td><td>Namespace ID[2]</td></tr> <tr><td>17</td><td>Namespace ID[3]</td></tr> <tr><td>18</td><td>Namespace ID[4]</td></tr> <tr><td>19</td><td>Namespace ID[5]</td></tr> <tr><td>20</td><td>Namespace ID[6]</td></tr> <tr><td>21</td><td>Namespace ID[7]</td></tr> <tr><td>22</td><td>Namespace ID[8]</td></tr> <tr><td>23</td><td>Namespace ID[9]</td></tr> <tr><td>24</td><td>Instance ID[0]</td></tr> <tr><td>25</td><td>Instance ID[1]</td></tr> <tr><td>26</td><td>Instance ID[2]</td></tr> <tr><td>27</td><td>Instance ID[3]</td></tr> <tr><td>28</td><td>Instance ID[4]</td></tr> <tr><td>29</td><td>Instance ID[5]</td></tr> <tr><td>30</td><td>Reserved</td></tr> <tr><td>31</td><td>Reserved</td></tr>	Namespace ID	0x0102030405060708090A	Instance ID	0x102030405060	RSSI@0m	0x00	LEN.	TYPE	VALUE	2	0x01	0x06	3	0x03	0xAAFE	23	0x16	0xAAFE00000102030405060708090A1020304050600000	02	Type: 0x01	03	Data: 0x06	04	Length: 0x03	05	Type: 0x03	06	Eddystone_UUID_LSB: 0xAA	07	Eddystone_UUID_MSB: 0xFE	08	Length: 0x17	09	Type: 0x16	10	Eddystone_UUID_LSB: 0xAA	11	Eddystone_UUID_MSB: 0xFE	12	Frame type UUID: 0x00	13	RSSI@0m: 0x00	14	Namespace ID[0]	15	Namespace ID[1]	16	Namespace ID[2]	17	Namespace ID[3]	18	Namespace ID[4]	19	Namespace ID[5]	20	Namespace ID[6]	21	Namespace ID[7]	22	Namespace ID[8]	23	Namespace ID[9]	24	Instance ID[0]	25	Instance ID[1]	26	Instance ID[2]	27	Instance ID[3]	28	Instance ID[4]	29	Instance ID[5]	30	Reserved	31	Reserved
	Namespace ID	0x0102030405060708090A																																																																															
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	02	Type: 0x01																																																																															
	03	Data: 0x06																																																																															
	04	Length: 0x03																																																																															
	05	Type: 0x03																																																																															
	06	Eddystone_UUID_LSB: 0xAA																																																																															
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	08	Length: 0x17																																																																															
	09	Type: 0x16																																																																															
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Table 1: Content structure of UID advertisement

2.2 Google Eddystone – URL

In terms of URL advertisement frame type, this is exactly complied with Google Eddystone regulations, and you can refer to below link for more information: <https://github.com/google/eddystone/tree/master/eddystone-url>

Below table described the detailed content structure and frame example of **URL** advertisement:

Eddystone - URL																					
Frame bytes	Frame format		Frame example																		
	01	Length: 0x02	Received frame : Eddystone - URL <table><tr><td>RSSI@0m</td><td>0x00</td></tr><tr><td>URL Scheme Prefix</td><td>0x00 = http://www.</td></tr><tr><td>Encoded URL</td><td>0x6D6F6B6F736D617274 = mokosmart (ASCII) 0x07 = .com (URL Expansion defined by Eddystone)</td></tr></table> Raw data: 0x0201060303AAFE1016AAFE1000006D6F6B6F736D61727407 Details: <table><tr><th>LEN.</th><th>TYPE</th><th>VALUE</th></tr><tr><td>2</td><td>0x01</td><td>0x06</td></tr><tr><td>3</td><td>0x03</td><td>0xAAFE</td></tr><tr><td>16</td><td>0x16</td><td>0xAAFE1000006D6F6B6F736D61727407</td></tr></table> LEN. - length of EIR packet (Type + Data) in bytes, TYPE - the data type as in https://www.bluetooth.org/en-us/specification/assigned-numbers/generic-access-profile OK	RSSI@0m	0x00	URL Scheme Prefix	0x00 = http://www.	Encoded URL	0x6D6F6B6F736D617274 = mokosmart (ASCII) 0x07 = .com (URL Expansion defined by Eddystone)	LEN.	TYPE	VALUE	2	0x01	0x06	3	0x03	0xAAFE	16	0x16	0xAAFE1000006D6F6B6F736D61727407
	RSSI@0m	0x00																			
	URL Scheme Prefix	0x00 = http://www.																			
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	LEN.	TYPE		VALUE																	
	2	0x01		0x06																	
	3	0x03		0xAAFE																	
	16	0x16		0xAAFE1000006D6F6B6F736D61727407																	
	02	Type: 0x01																			
	03	Data: 0x06																			
	04	Length: 0x03																			
	05	Type: 0x03																			
	06	Eddystone_UUID_LSB: 0xAA																			
	07	Eddystone_UUID_MSB: 0xFE																			
	08	Length: ≤ 0x17																			
	09	Type: 0x16																			
	10	Eddystone_UUID_LSB: 0xAA																			
	11	Eddystone_UUID_MSB: 0xFE																			
	12	Frame type UUID: 0x10																			
	13	RSSI@0m: 0x00																			
	14	URL Scheme Prefix: 0x00																			
	15	Encoded URL[0]																			
	16	Encoded URL[1]																			
	17	Encoded URL[2]																			
	18	Encoded URL[3]																			
	19	Encoded URL[4]																			
	20	Encoded URL[5]																			
	21	Encoded URL[6]																			
	22	Encoded URL[7]																			
	23	Encoded URL[8]																			
	24	Encoded URL[9]																			
25	Encoded URL[10]																				
26	Encoded URL[11]																				
27	Encoded URL[12]																				
28	Encoded URL[13]																				
29	Encoded URL[14]																				
30	Encoded URL[15]																				
31	Encoded URL[16]																				

Table 2: Content structure of URL advertisement

2.3 Google Eddystone – TLM (Unencrypted)

In terms of TLM advertisement frame type, this is exactly complied with Google Eddystone regulations, and for more details you can refer to below link: <https://github.com/google/eddystone/blob/master/eddystone-tlm/tlm-plain.md>

Below table described the detailed content structure and frame example of **TLM** advertisement:

Eddystone - TLM																									
Frame bytes	Frame format		Frame example																						
	01	Length: 0x02	Received frame : Eddystone - TLM <table><tr><td>TLM version</td><td>0x00 = Unencrypted</td></tr><tr><td>Battery voltage</td><td>0x0C87 = 3207mV</td></tr><tr><td>Beacon temperature</td><td>0x0D80 = 13.5°C (8.8 fixed-point notation)</td></tr><tr><td>ADV_CNT (1 time/unit)</td><td>0x00000659 = 1625 times Remark: running count of advertisement frames of all types emitted by the beacon since power-up or reboot</td></tr><tr><td>SEC_CNT (0.1s/unit)</td><td>0x00003F6F = 1623.9s (00:27:03.900s) Remark: It is a 0.1 second resolution counter that represents time since beacon power-up or reboot</td></tr></table> Raw data: 0x0201060303AAFE1116AAFE20000C870D800000065900003F6F Details: <table><tr><th>LEN.</th><th>TYPE</th><th>VALUE</th></tr><tr><td>2</td><td>0x01</td><td>0x06</td></tr><tr><td>3</td><td>0x03</td><td>0xAAFE</td></tr><tr><td>17</td><td>0x16</td><td>0xAAFE20000C870D800000065900003F6F</td></tr></table> LEN. - length of EIR packet (Type + Data) in bytes, TYPE - the data type as in https://www.bluetooth.org/en-us/specification/assigned-numbers/generic-access-profile OK Eddystone - TLM field	TLM version	0x00 = Unencrypted	Battery voltage	0x0C87 = 3207mV	Beacon temperature	0x0D80 = 13.5°C (8.8 fixed-point notation)	ADV_CNT (1 time/unit)	0x00000659 = 1625 times Remark: running count of advertisement frames of all types emitted by the beacon since power-up or reboot	SEC_CNT (0.1s/unit)	0x00003F6F = 1623.9s (00:27:03.900s) Remark: It is a 0.1 second resolution counter that represents time since beacon power-up or reboot	LEN.	TYPE	VALUE	2	0x01	0x06	3	0x03	0xAAFE	17	0x16	0xAAFE20000C870D800000065900003F6F
	TLM version	0x00 = Unencrypted																							
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	Beacon temperature	0x0D80 = 13.5°C (8.8 fixed-point notation)																							
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	LEN.	TYPE		VALUE																					
	2	0x01		0x06																					
	3	0x03		0xAAFE																					
	17	0x16		0xAAFE20000C870D800000065900003F6F																					
	02	Type: 0x01																							
	03	Data: 0x06																							
	04	Length: 0x03																							
	05	Type: 0x03																							
	06	Eddystone_UUID_LSB: 0xAA																							
	07	Eddystone_UUID_MSB: 0xFE																							
	08	Length: 0x11																							
	09	Type: 0x16																							
	10	Eddystone_UUID_LSB: 0xAA																							
	11	Eddystone_UUID_MSB: 0xFE																							
	12	Frame type UUID: 0x20																							
	13	TLM version: 0x00																							
	14	Battery voltage (1mV)_LSB																							
	15	Battery voltage (1mV)_MSB																							
	16	Beacon temperature (0.1°C)_LSB																							
	17	Beacon temperature (0.1°C)_MSB																							
	18	ADV_CNT[0]																							
	19	ADV_CNT[1]																							
	20	ADV_CNT[2]																							
	21	ADV_CNT[3]																							
	22	SEC_CNT[0]																							
	23	SEC_CNT[1]																							
24	SEC_CNT[2]																								
25	SEC_CNT[3]																								
26	Not used																								
27	Not used																								
28	Not used																								
29	Not used																								
30	Not used																								
31	Not used																								

Table 3: Content structure of TLM advertisement

2.4 Customized frame – “iBeacon”

This *Customized iBeacon* advertisement frame is combined with two parts, of which one is standard APPLE iBeacon format which comply with APPLE iBeacon regulations, another one is a customized scan response packet.

What the purpose of adding extra scan response packet is just displaying more information on iOS APP, as APP cannot get the detail information from APPLE iBeacon advertisement due to iOS system restrictions. For APPLE iBeacon regulations, you can refer to the link: <https://developer.apple.com/ibeacon/Getting-Started-with-iBeacon.pdf>.

Below table described the detailed content structure and frame example of **Customized-iBeacon** advertisement:

MOKO Customized - iBeacon			
Frame bytes		Standard iBeacon packet	Scan response packet
	01	Length: 0x02	Length: 0x02
	02	Type: 0x01	Type: 0x0A
	03	Data: 0x06	Tx Power: 0x00
	04	Length: 0x1A	Length: 0x1A
	05	Type: 0xFF	Type: 0x16
	06	Apple CIN_LSB: 0x4C	Service data UUID_LSB: 0xAB (MOKO defined)
	07	Apple CIN_MSB: 0x00	Service data UUID_MSB: 0xFE (MOKO defined)
	08	Beacon type: 0x02	Frame type: 0x50 (MOKO defined)
	09	Data size: 0x15	RSSI@1m: 0xC5
	10	UUID[0]	ADV interval: 0x0A
	11	UUID[1]	UUID[0]
	12	UUID[2]	UUID[1]
	13	UUID[3]	UUID[2]
	14	UUID[4]	UUID[3]
	15	UUID[5]	UUID[4]
	16	UUID[6]	UUID[5]
	17	UUID[7]	UUID[6]
	18	UUID[8]	UUID[7]
	19	UUID[9]	UUID[8]
	20	UUID[10]	UUID[9]
	21	UUID[11]	UUID[10]
	22	UUID[12]	UUID[11]
	23	UUID[13]	UUID[12]
	24	UUID[14]	UUID[13]
	25	UUID[15]	UUID[14]
	26	Major[0]	UUID[15]
	27	Major[1]	Major[0]
	28	Minor[0]	Major[1]
	29	Minor[0]	Minor[0]
	30	RSSI@1m: 0xC5	Minor[0]
	31	Not used	Not used

Table 4: Content structure of Customized – iBeacon advertisement

Received frame : MOKO Customized - iBeacon	
UUID	0x0001020304050607080910A0B0C0D0E0F
Major	0x0001
Minor	0x0001
RSSI@1m	0xC5 = -59dBm
Tx Power	0x00 = 0dBm
ADV interval (100ms/unit)	0x0A = 1000ms

Raw data:

```

0x0201061AFF4C00021500010203040
5060708090A0B0C0D0E0F00010001C
5020A001A16ABFE50C50A000102030
405060708090A0B0C0D0E0F0001000
1

```

Details:

	LEN.	TYPE	VALUE
APPLE-iBeacon	2	0x01	0x06
	26	0xFF	0x4C000215000102030405060708090A0B0C0D0E0F00010001C5
Scan Response Packet	2	0x0A	0x00
	26	0x16	0xABFE50C50A000102030405060708090A0B0C0D0E0F00010001

LEN. - length of EIR packet (Type + Data) in bytes,
TYPE - the data type as in <https://www.bluetooth.org/en-us/specification/assigned-numbers/generic-access-profile>

OK

Standard iBeacon packet field

Tx Power

Scan response packet field

Figure 1: Example content structure of Customized – iBeacon advertisement

2.5 Customized frame – “Tag info”

This “Tag info” advertisement frame is customized by MOKO and also combined with two parts which is “advertising packet” and “scan response packet”.

- **ADV_IND (advertising packet)** ----Mainly to express the tag ID, sensor status, trigger event counts, battery voltage and acceleration data.
- **SCAN_RSP (scan response packet)** ----Mainly to express the device name.

Below table described the detailed content structure and frame example of **Customized-Tag info** advertisement:

MOKO Customized - Tag info			
Frame bytes		Advertisement packet	Scan response packet
	01	Length: 0x02	Length: ≤ 0x15
	02	Type: 0x01	Type: 0x09
	03	Data: 0x06	Device name[0]
	04	Length: 0x18	Device name[1]
	05	Type: 0x16	Device name[2]
	06	Service data UUID_LSB: 0x01 (MOKO defined)	Device name[3]
	07	Service data UUID_MSB: 0xEA (MOKO defined)	Device name[4]
	08	Frame type UUID: 0x80 (MOKO defined)	Device name[5]
	09	Sensor status (Magnetic/Motion)	Device name[6]
	10	MAGNET_CNT_LSB	Device name[7]
	11	MAGNET_CNT_MSB	Device name[8]
	12	MOTION_CNT_LSB	Device name[9]
	13	MOTION_CNT_MSB	Device name[10]
	14	Accel.Dat_ X axis (mg) LSB	Device name[11]
	15	Accel.Dat_ X axis (mg) MSB	Device name[12]
	16	Accel.Dat_ Y axis (mg) LSB	Device name[13]
	17	Accel.Dat_ Y axis (mg) MSB	Device name[14]
	18	Accel.Dat_ Z axis (mg) LSB	Device name[15]
	19	Accel.Dat_ Z axis (mg) MSB	Device name[16]
	20	Reserved	Device name[17]
	21	Reserved	Device name[18]
	22	Reserved	Device name[19]
	23	Reserved	Not used
	24	Battery voltage (1mV)_LSB	Not used
	25	Battery voltage (1mV)_MSB	Not used
	26	Tag ID[0]	Not used
	27	Tag ID[1]	Not used
	28	Tag ID[2]	Not used
	29	Tag ID[3]	Not used
	30	Tag ID[4]	Not used
	31	Tag ID[5]	Not used

Table 5: Content structure of Customized – Tag info advertisement

Received frame : MOKO Customized - Tag info															
Sensor status	[Bit0]: Magnetic sensor working status	0x04 = 0000 0100'b	[Bit0]: 0 = Magnet is approaching												
	[Bit1]: Motion sensor working status		[Bit1]: 0 = Device is in static												
	[Bit2]: Motion sensor equipped status		[Bit2]: 1 = Motion sensor is equipped in the device												
	[Bit7-Bit2]: Reserved		[Bit7-Bit3]: Reserved												
MAGNET_CNT		0x0000 = 0 (Times of magnetic triggered)													
MOTION_CNT		0x0002 = 2 (Times of motion triggered)													
Accel.Data_X axis		0x003A = 58mg													
Accel.Data_X axis		0x0076 = 118mg													
Accel.Data_X axis		0x03A7= 935mg													
Battery voltage (1mV/unit)		0x0CCC = 3276mV													
Tag ID		0x000001													
Device name		0x4D4B20546167 = MK Tag													
Advertisement packet filed Device name Raw data: 0x020106181601EA800400000002003 A007603A7FFFFFFFFF0CCC000001070 94D4B20546167 Details: <table><thead><tr><th>LEN.</th><th>TYPE</th><th>VALUE</th></tr></thead><tbody><tr><td>2</td><td>0x01</td><td>0x06</td></tr><tr><td>24</td><td>0x16</td><td>0x01EA800400000002003A0076 03A7FFFFFFFFF0CCC000001</td></tr><tr><td>7</td><td>0x09</td><td>0x4D4B20546167</td></tr></tbody></table> LEN. - length of EIR packet (Type + Data) in bytes, TYPE - the data type as in Generic Access Profile.pdf OK				LEN.	TYPE	VALUE	2	0x01	0x06	24	0x16	0x01EA800400000002003A0076 03A7FFFFFFFFF0CCC000001	7	0x09	0x4D4B20546167
LEN.	TYPE	VALUE													
2	0x01	0x06													
24	0x16	0x01EA800400000002003A0076 03A7FFFFFFFFF0CCC000001													
7	0x09	0x4D4B20546167													

Figure 2: Content structure of Customized – Tag info advertisement

3. “MOKO TAG” APP user guidance

Well, you have already familiar with the advertisement format and well to know how to parse the content in the advertisement, now we will guide you to know the “MOKO TAG” APP and let’s do some configuration on the Tag. **For the latest APP, please search “MOKO TAG” and download from Google play store or Apple store.**

This user guidance for “MOKO TAG” app is based on Android/iOS APP version 1.0.0 or above, and the content of this user guidance are subject to change without prior notice for further improvement.

3.1 Discover your tag

The smartphone with “MOKO TAG” APP (Android or iOS) installed can be as a *Bluetooth Center* to scan and discover the advertisement packets of Bluetooth Low Energy (BLE) Beacons, and it will provide multiple strategies to filter the Bluetooth packets and that can help you to capture the required BLE peripherals directly and quickly.

3.1.1 Scanning your tag

When enter into the “**SCANNING PREVIEW**” interface, the *SCANNER* will keep scanning continuously and list all surrounding advertising Bluetooth LE peripherals which integrated with “**BXP-T**” series firmware, unless stop scanning manually by pressing scanning button “↻”.

In this *SCANNER* interface, you will see device advertisement preview to recognize your Beacon device. To distinguish specific type of devices separately, only Beacon devices with MOKO “**BXP-T**” series firmware can be shown in the APP preview interface and others will be filtered automatically.

Below picture shows the example scanning preview of MOKO Tag series products.

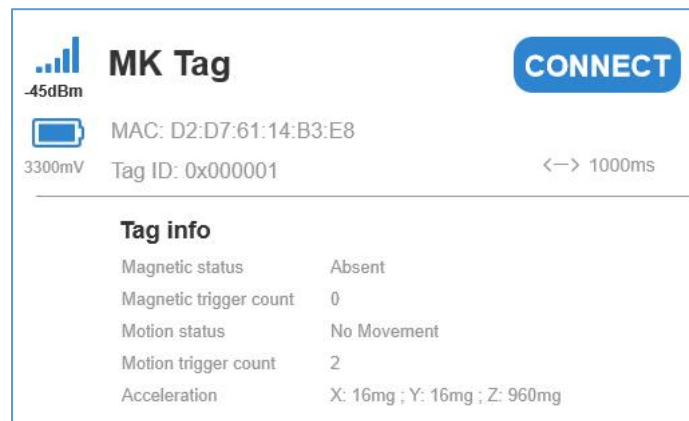
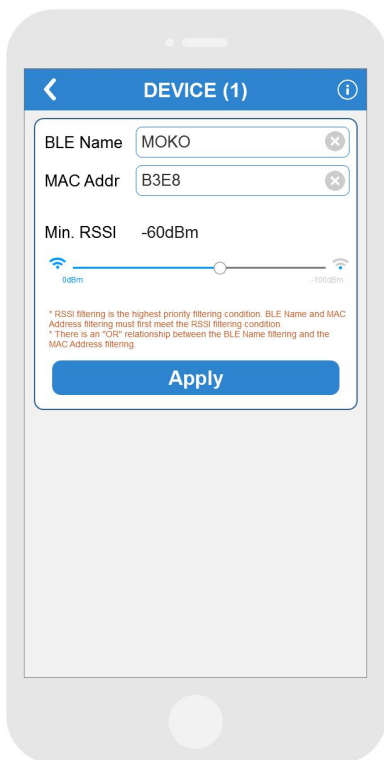


Figure 3: Example scanning preview of MOKO TAG

3.1.2 Filtering your tag

To find your tag quickly, we have developed multiple strategies to filter among mass Beacon devices, users may narrow your device list by BLE name, MAC address and RSSI value through “**Edit Filter**” option.



BLE Name	Allow to input 1-20 characters. Example: Only Beacon devices whose BLE name contain “MOKO” will be displayed in SCANNER interface. NOTE: Space effective, and case insensitive as well.
Mac Addr	Allow to input 1-6 bytes (HEX). Example: Only Beacon devices whose Mac address contain “B3:E8” will be displayed in SCANNER interface. NOTE: There is no need to input colon between MAC address in search filter column, and case insensitive as well.
Min.RSSI	Only the advertisement packets with RSSI not less than the Min.RSSI value will be displayed in SCANNER interface, and configuration range is -100dBm to 0dBm. Example: When setting the Min.RSSI value to -70, the ADV packets with RSSI of -70dBm or greater than -70dBm will be list in SCANNER interface.

Figure 4: Configurations of scanning filter settings

NOTE:

- ① RSSI is the highest priority filtering condition. It means that filtered device must meet the RSSI filtering condition.
- ② There is an “OR” relationship between BLE Name and Mac Addr. It means that filtered device just need meet one of the filtering conditions.

3.2 Connect with your tag

For the Tag with “BXP-T” series firmware, it will display “**CONNECT**” button in the “**SCANNING PREVIEW**” interface if the status is connectable and you can follow below steps to connect with your tag:

Step 1: Press “**CONNECT**” button.

Step 2: Input connection password and then press “**OK**” button, Green LED will blink 2 times rapidly again.



Notice – Default password in MOKO standard firmware is “Moko4321”.

3.3 Know your tag’s advertising logic

Before setting up your tag’s advertisement, you may need to understand the basic advertising logic and working flowchart of the tag. With the integration of ultra-long advertising interval (standby duration) feature, in order to guarantee the efficiency of connections based on ultra-long advertising interval mode, we have integrated a constant advertising cycle which is called configuration period when device boot up.

The configuration period will last for 30 seconds and then device will start working (advertising period) by following the rules you preset in the advertisement SLOTS.

Remark: If standby duration is set into 0, then adv duration value will be invalid and device will keep advertising no matter what adv duration value you set.

Overall advertising logic		
Advertising Period	Device status	Default Configurations
Configuration period	Connectable	Adv interval – 1000ms (non-configurable) Adv duration – 30s (non-configurable) <i>*If you have modification requirements on this, please contact with sales for support.</i>
Advertising period	Connectable	Adv interval – 1000ms (configurable) Adv duration – 10s (configurable) Standby duration – 0s (configurable)

Table 6: Different advertising periods in overall advertising logic

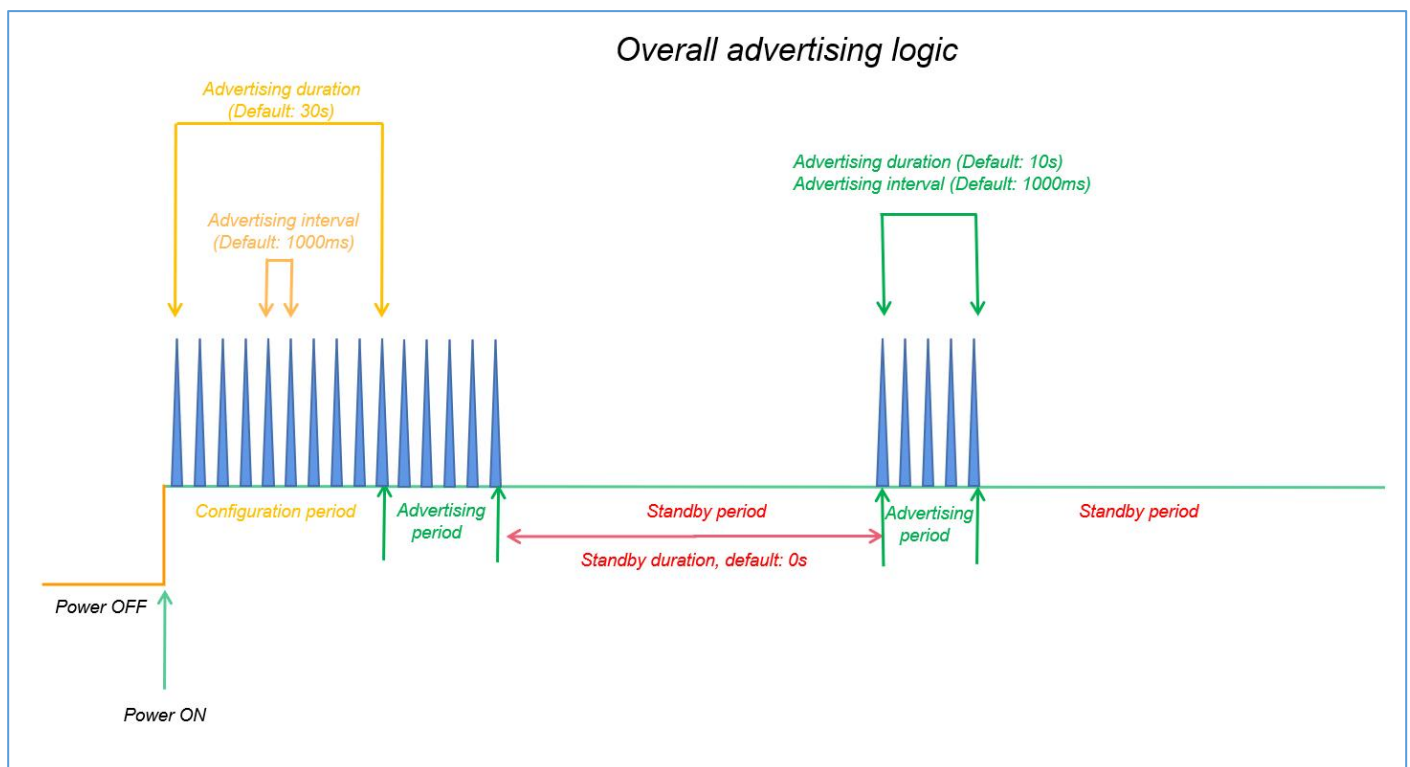


Figure 5: Overall advertising logic

3.4 How to apply to your scenarios?

To expand the usage scenarios of “MOKO TAG” series products in asset management, we have implemented some specific features to make it preferentially and exclusively adapted to various asset tracking situations, such as movement-based tracking mode, asset presence detection and ultra-long advertising interval etc.

Here we will introduce some typical scenarios for your reference based on practical cooperations with end users. **In order to ensure the feasibility of settings for all users in different application scenarios, we do not directly define the mode switch but need users to configure relevant parameters to achieve this.** Under this circumstance, we have also made corresponding flowchart to elaborate the details and guide you how to set this in the APP interface.

➤ Presence detection mode

➤ Movement-based dynamic tracking mode

3.4.1 Presence detection mode

This presence detection mode is commonly used for fixed assets, such as furniture/desks/photocopiers in the office, or medical equipments/beds in the hospital. In these practical scenarios, you will only need to know it is in right place rather than the real-time positions, so many end users prefer to keep device broadcasting every 1 hour or more.

Below described the typical configurations and working logical of presence detection mode:

Typical scenario 1 – Presence detection		
items	Descriptions	Configurations
Working logical	Device will broadcast 10s with 1000ms advertising interval every 1 hour.	-
Typical configurations	Advertising interval	1000ms
	Advertising duration	10s
	Standby duration	3600s
Motion sensor	Not necessary	-
Suggested scenarios	Smart office – Furniture/desks/photocopiers tracking Healthcare – Medical equipment/beds tracking	-

Table 7: Typical parameters configuration of presence detect mode

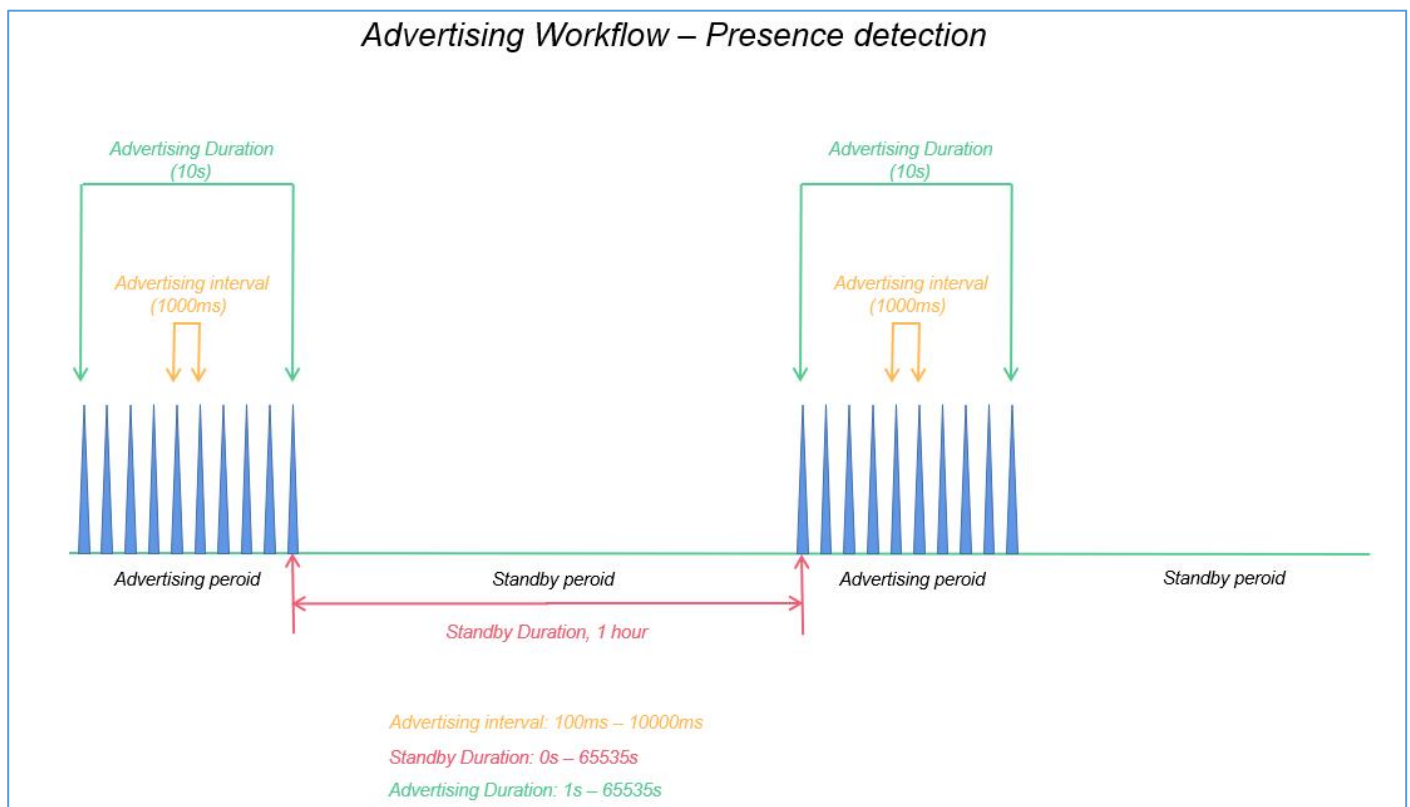


Figure 6: Advertising workflow of typical presence detection mode

3.4.2 Movement-based dynamic tracking mode

This dynamic tracking mode is commonly used for RTLS application scenarios, such as packaging box/container/pallet in the warehouse, frequently used IV infusion pumps/wheelchairs in the hospital, or shopping carts in the market. In these practical scenarios, you must need to know its real-time positions rather than simple presence status, and in order to save the battery consumption when assets in no use status (detected by motion sensor), you can lower the

advertisement report frequency when assets in long-terms idle status. Based on these demands, many clients prefer the movement-based dynamic tracking mode.

Below described the typical configurations and working logical of dynamic tracking mode:

Typical scenario 2 – Movement-based dynamic tracking mode		
items	Descriptions	Configurations
Working logical	Device will broadcast rapidly when in use (moving) and broadcast slowly when in no use (static)	-
Typical configurations SLOT 1	Advertising frame	iBeacon
	Advertising interval	2500ms
	Advertising duration	10s
	Standby duration	0s
	Trigger – motion detection	Stop advertising after device keep static for 30s
Typical configurations SLOT 2	Advertising frame	iBeacon
	Advertising interval	1000ms
	Advertising duration	5s
	Standby duration	300s
	Trigger – motion detection	Start advertising for 0s after device keep static for 30s
Motion sensor	Necessary	
Suggested scenarios	Warehouse – Packaging box/pallet tracking Healthcare – IV infusion pump/wheelchairs tracking Entertainment – Shopping cart tracking	-

Table 8: Typical parameters configuration of movement-based dynamic tracking mode

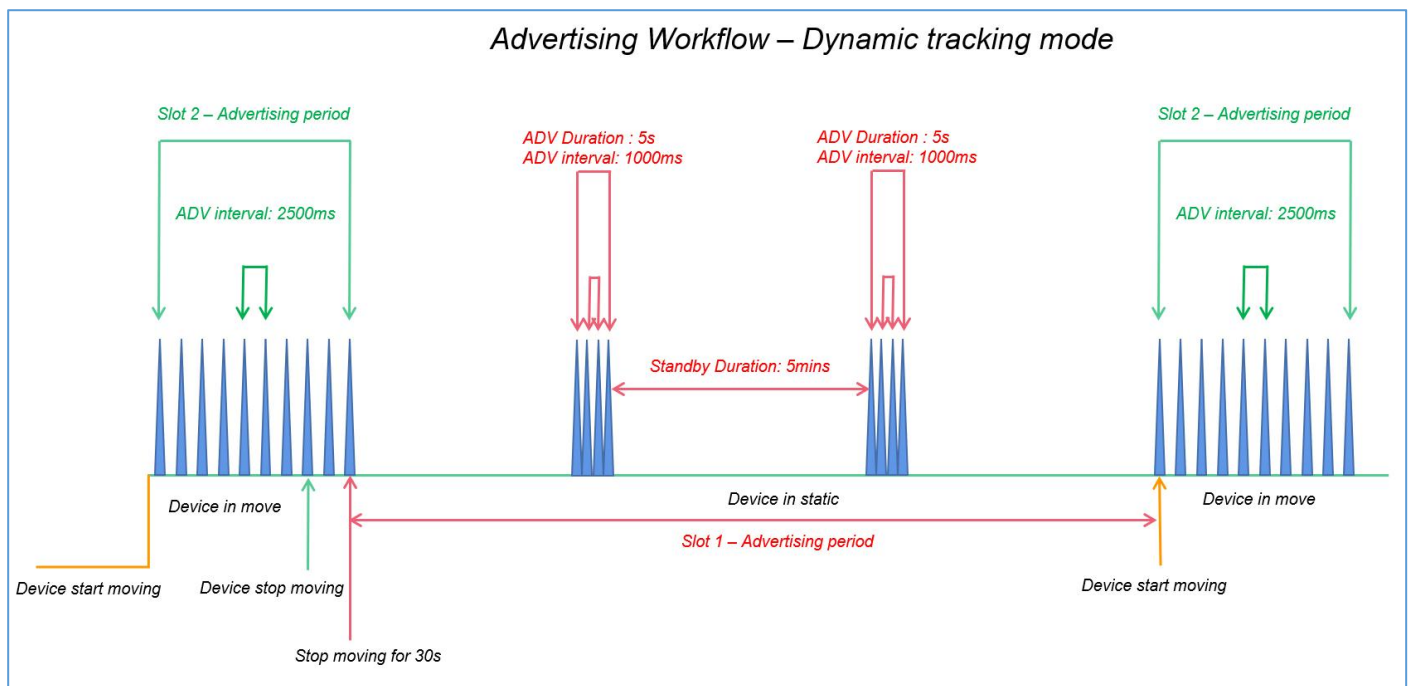


Figure 7: Advertising workflow of typical dynamic tracking mode

3.5 Access to sensor configurations

3.5.1 Motion sensor (3-axis accelerometer)

Regarding of MOKO TAG series product, currently we have adopted motion sensor for “Movement-based dynamic tracking” functions which link to “[Chapter 3.4.2 : Movement-based dynamic tracking mode](#)” for your reference, and you can access to “**SETTING >>>Sensor configurations>>> 3-axis accelerometer**” interface for further configurations.

Remark: Configuration interface will be shown only when device equipped with 3-axis accelerometer sensor.

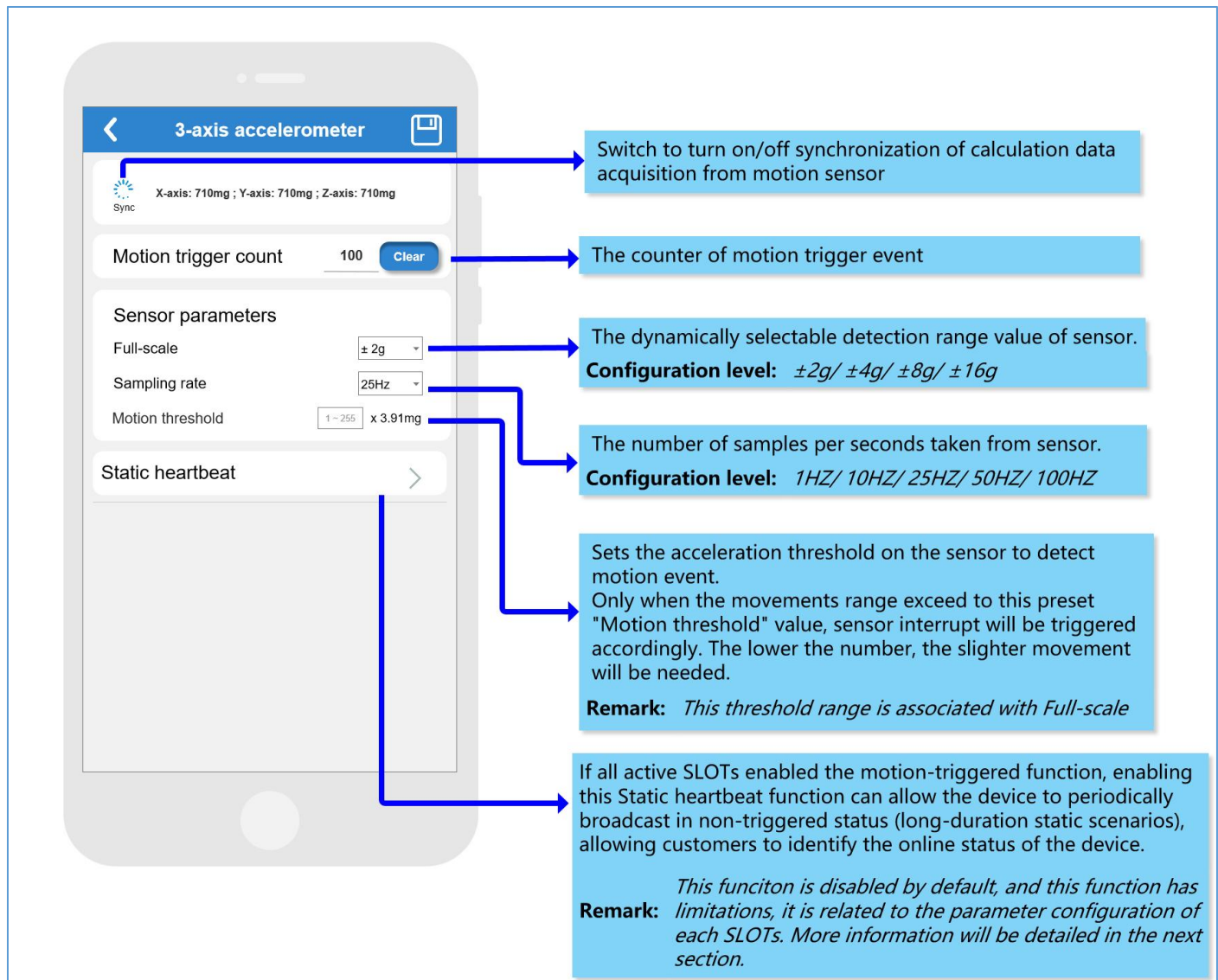


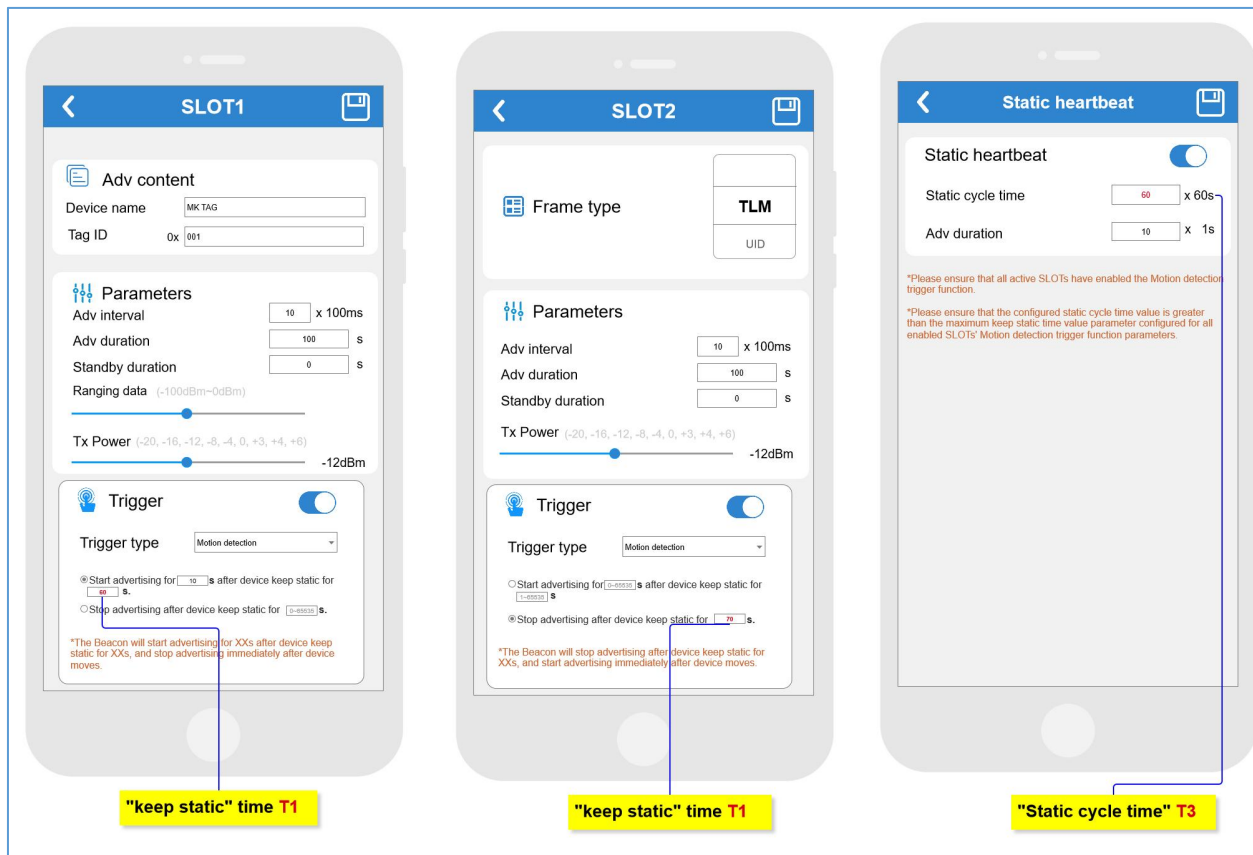
Figure 8: Configurations page of motion sensor

3.5.1.1 Static heartbeat

This feature is mainly applicable in situations where the device will remain stationary for a very long time. After reaching the fixed static duration condition, the device will start a fixed duration broadcast to ensure that terminal users can get the online status of the device, it just like the device's "heartbeat broadcast". This feature is a global configuration for the mobile trigger function and it is disabled by default.

If you want to enable this feature, you need to understand the prerequisites for enabling it. This is very important, otherwise you will not be able to properly use it.

1. The "Static heartbeat" feature requires that all active Slots have enabled the **Motion detection** function.
2. The configuration of the Static heartbeat function's **Static cycle time(T3)** has a constraint condition: First, you need comparing and calculating the maximum value of the **"keep static" time (T1)** configured for the **Motion detection** function parameters in all activated Slots, the parameter **T3** needs to be greater than this maximum value.
3. Once the **Static heartbeat** function is enabled, the **"Static heartbeat"** advertising type will be broadcast according to the first enabled slot's broadcast type and broadcast interval, and the broadcast duration is the duration of the **Adv duration** time configured on the page for the **Static heartbeat** function.



3.5.2 Magnetic sensor (hall sensor)

Currently we have adopted the same IO in the hardware design for magnetic sensor and mechanical button (if device have), so we have designed a function switch for multiple use as a matter of course.

Feature 1: Magnetic sensor for power on/off only.

Feature 2: Magnetic sensor for power on and trigger event detection only.

For more details or acquiring the real-time magnetic sensor working status, you can access to “**SETTING >>>Sensor configurations>>> Hall sensor**”.

Remark: This feature is only applicable for the device which equipped with magnetic sensor.

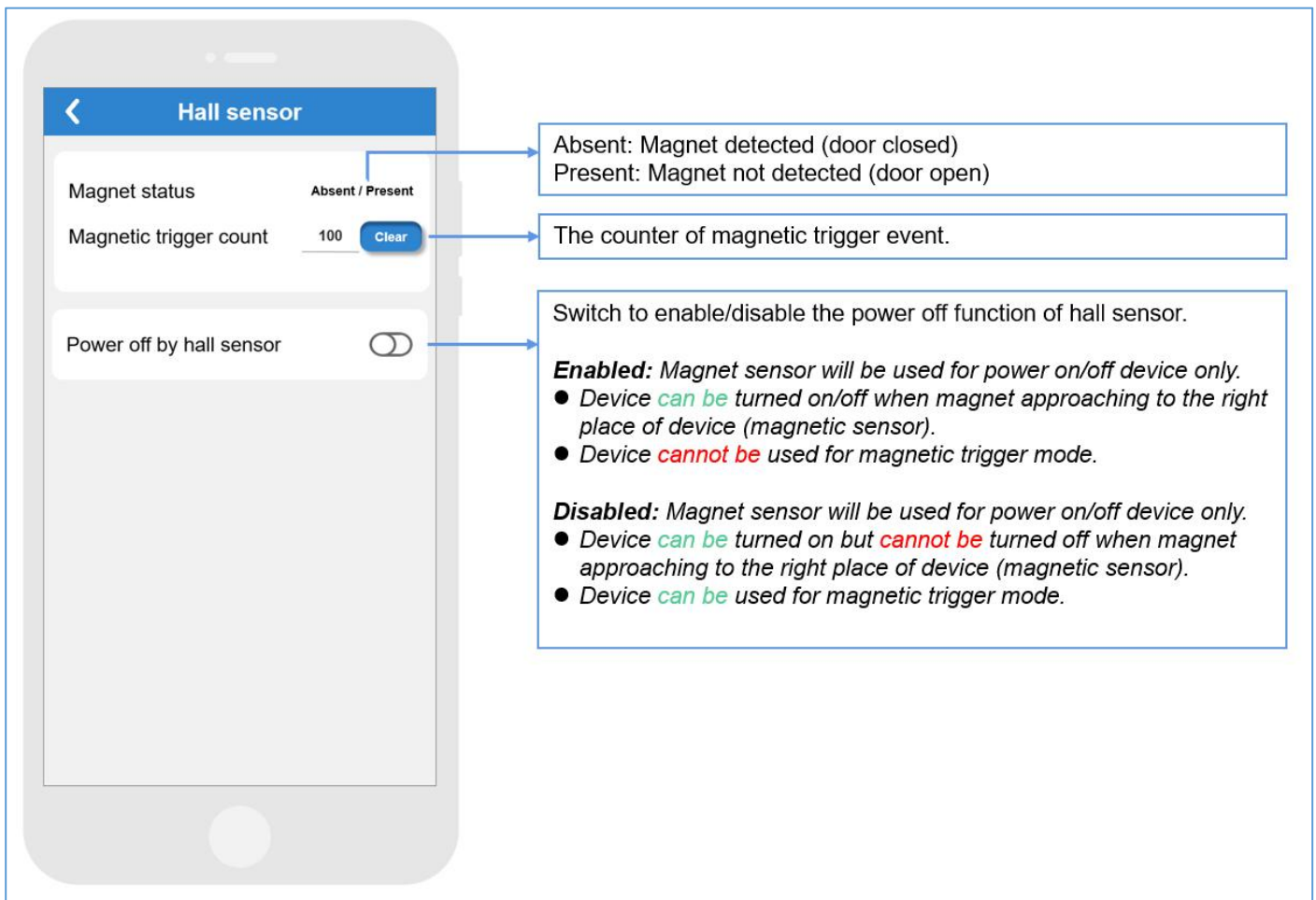


Figure 10:Configurations page of magnetic sensor

3.6 How to enable “Long range” features?

In order to enhance the transmission range of the tag and expand the application scenarios as well as reduce the gateway deployment costs, we have optimized the firmware feature and integrated with coded PHY mode. Based on current firmware, “1M PHY – Legacy” and “Coded PHY – Long Range” are supported already, and you can access to “**SETTING >>>ADV mode**” and adopt the right one to your application scenarios.

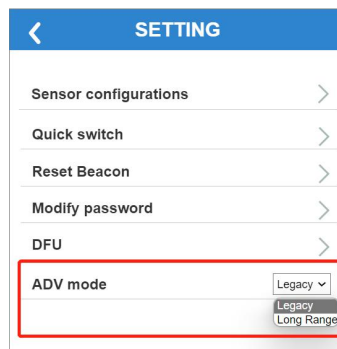


Figure 11: How to enable “Long Range” mode?

➤ **Legacy** --- 1M PHY

➤ **Long Rang** --- Coded PHY

Normally the transmission range in Coded PHY mode will be multiple times of transmission range in 1M PHY, and the advertisement payload format/content will be remaining with only the changes of advertisement type due to BLE 5.0 new specifications.

Here we have listed the ADV type difference between 1M PHY and Coded PHY. And for more introduction you can refer to link: <https://www.bluetooth.com/bluetooth-resources/intro-to-bluetooth-low-energy-coded-phy/>

Frame type	PHYs			
	LE 1M		LE Coded	
	Connectable status	Non-connectable status	Connectable status	Non-connectable status
iBeacon	ADV_IND + SCAN_RSP	ADV_SCAN_IND	ADV_EXT_IND + AUX_ADV_IND	
Eddystone – URL/TLM/UID	ADV_IND	ADV_NONCONN_IND	ADV_EXT_IND + AUX_ADV_IND	
Tag info	ADV_IND + SCAN_RSP	ADV_SCAN_IND	ADV_EXT_IND + AUX_ADV_IND	

Table 9: ADV type difference between 1M and Coded PHY

3.7 Quick switch for basic features

In current firmware, we have designed multiple function switch for some basic functions. Based on that, you can disable/enable these features through the switch, thus adapting to different use cases.

You can enable/disable these function switch in **“Setting >>> Quick switch”** section, and below shows the **“Quick switch”** interface as well as relevant feature descriptions.

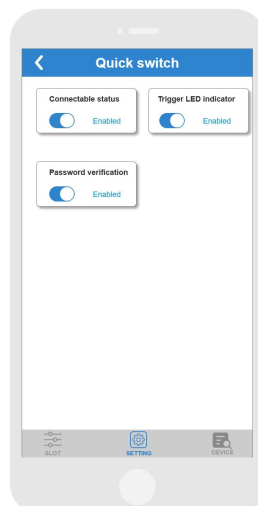


Figure 12: Configurations page of Quick switch

3.7.1 Switch your Tag to unconnectable status

In some cases, you will need to disable the tag's connectable property, to let it work just as a location tag. So you can disable the connectable status in "**Setting >>> Quick switch >>> Connectable status**". After feature disabled, device won't be connected by master BLE device anymore unless reset your tag.



Figure 13: Switch your Tag to unconnectable status

3.7.2 Disable password verification

You can go to "**Setting >> Quick switch >> Password verification**" to disable password verification. Once password verification be disabled, there will be no need to input password when you connect with your tag.

Remark: If password verification feature was disabled, you will not be able to change password or reset Beacon due to security process restrictions.



Figure 14: How to disable password verification?

3.7.3 Disable trigger LED indicator

If you preset the motion trigger or magnetic trigger mechanism, the LED will blink for 1 time when it is triggered. Considering the invisibility in some specific scenarios or the save of battery consumption, you can access to "**Setting >> Quick switch >> Trigger LED indicator**" to disable the LED notification in trigger mode.



Figure 15: How to disable trigger LED indicator?

3.8 Factory reset your Beacon

3.8.1 Remote reset

In some cases that mechanical button function should be disabled, to avoid intentional reset situations. Under this circumstance we will need remote reset function for backup. You can remote reset your tag by just connect with your tag and tapping “**Setting >>> Reset Beacon**” option, then it will give prompt to you to confirm the reset operations.

Remark: Remote reset operations will not change the connection password to factory reset status. If you want to totally reset the Beacon with password restore as well, then you may need to do hardware reset operations.

3.8.2 Hardware reset

In order to avoid the stuck in the dilemma of not being able to restore factory settings due to the hardware design without physical buttons, we have provided a universal operation flow to execute the hardware reset.

Hardware reset operation flow:

- 1) Remove the cover and the battery.
- 2) Put the battery in until you see the LED keep blinking for 2 seconds. This LED blinking means that the battery is connected successfully.
- 3) When the LED blinking finishes, put the battery in again within 5 seconds, and then repeat the process four more times, removing and replacing the battery. You should see a LED blinking each time you put the battery in, for a total of five blinks. The fifth blinking is different from the previous four and indicates that the device is factory reset successfully.

1st – 4th : LED keep blinking for 2s. This indicates the battery is connected successfully.

5th : LED keep solid for 2s. This indicates the device is factory reset successfully.

- 4) Replace the cover by aligning the battery room and press down on the cover.

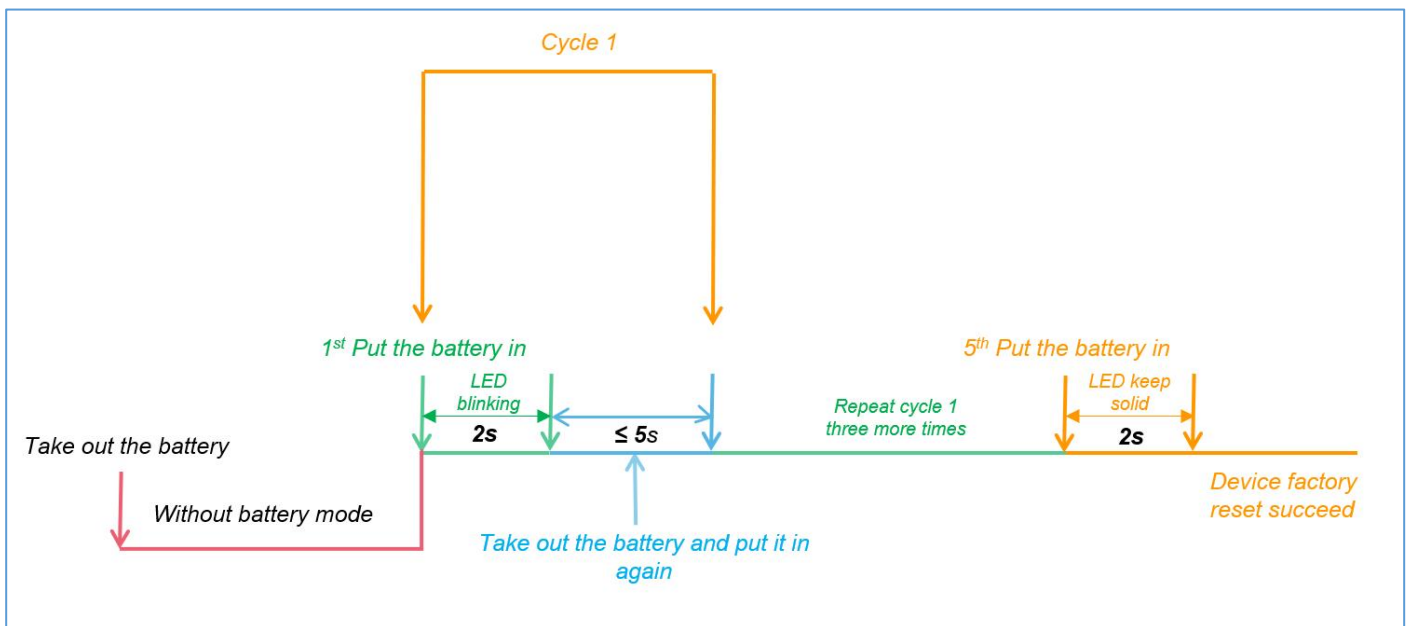


Figure 16: How to hardware reset you Tag without physical buttons?

3.9 Modify “Connection password”

In MOKO firmware, password verification is ON by default and “Moko4321” was set as default connection password. You can modify your Beacon’s connection password by just tapping “**DEVICE >>> modify password**” option, and then input the password as you want.

Remark: Connection password should not exceed 16 characters in length.

3.10 Remote control Beacon LED

In some cases when you need to find your Beacon, then you can use this “**Remote reminder**” functions. You can go to “**SETTING >>> Remote reminder**” and then set the remote remind notification type, then just tap “**Remind**” button, Beacon LED will start blinking according to the preset parameters.

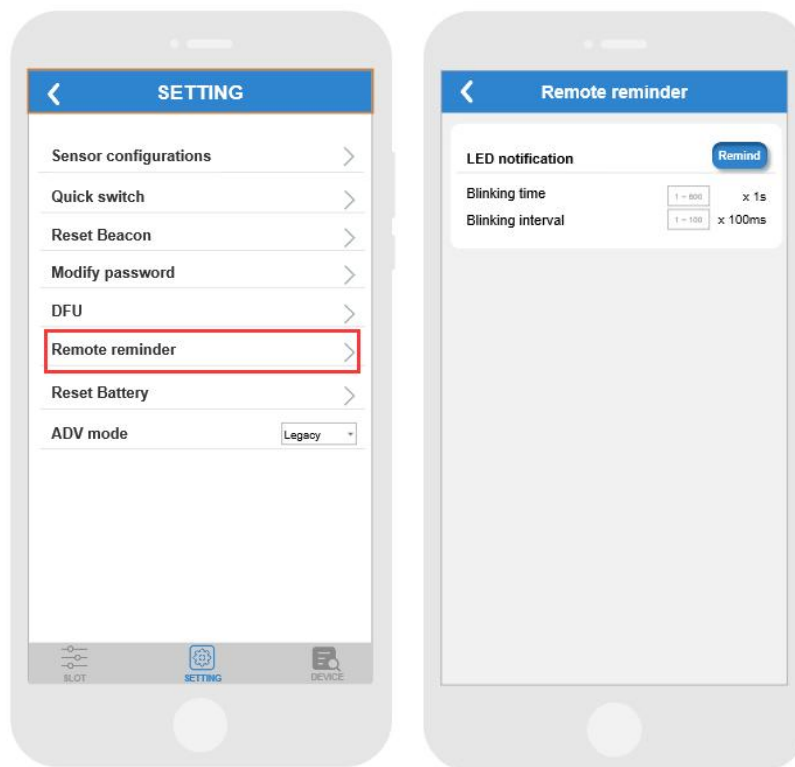


Figure 17: Remote reminder page

In “**Remote reminder**” page, there have two parameters that need to be set.

- **Blinking time.** ----- LED blinking duration after click the “**Remind**” button in the app. Configuration range: 1 – 6000 (Unit: 100ms)
- **Blinking interval.** ----- LED blinking interval. Configuration range: 1 – 100 (Unit: 100ms)

After configuration, you can control the LED of the beacon to blink by click the “**Remind**” button

Remark: This operation is a one-time operation. Each time you click the Remind button, the LED will only be executed once based on the current configuration. In addition, please make sure your device is equipped with LED before performing this operation

3.11 Reset battery

As some of our products use lithium-ion batteries, the calculation method of this type of battery is different from that of button batteries. Therefore, for products using lithium-ion batteries, this option will be displayed in the APP. After changing the battery, you need to click this option to make the device to recalculate the battery power.

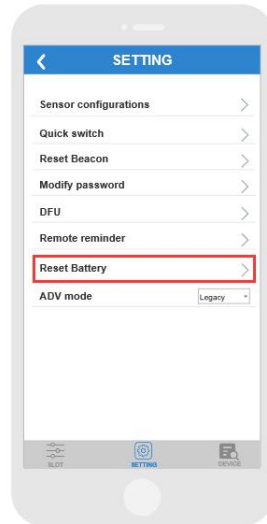


Figure 18: Reset battery instruction

Remark: If your product is the lithium sub-battery version, after replacing the battery, you need to connect the device and use this function to reset the battery of the device, otherwise, the battery power will not display the correct value. In addition, the button battery version of the product will not show this option box in the APP

3.12 General information

In some cases, you will need to know the firmware version of your Tag, or product model and etc. You can access to “**Device >>> Device info**” interface to check and confirm the essential information.

Remark: In customization firmware, that information can also be edit as customer required.

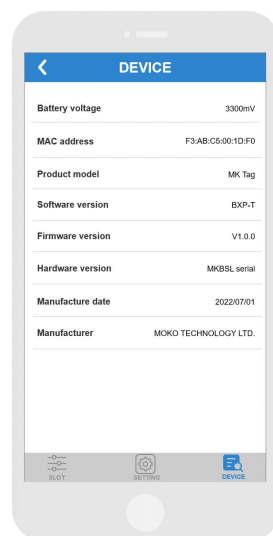


Figure 19: Device information page

3.13 Upgrade your Tag's firmware

The tag can realize OTA firmware update through DFU operations. Regarding of different import methods for android app and iOS app, here we will describe the detailed steps separately.

3.13.1 Android – MOKO APP DFU instructions

Step 1: Copy the upgrade package to a folder in the root directory of the smartphone.

Step 2: **Connect** with device and go to “**SETTING**” interface, press “**DFU**” option and then file manager route of smartphone will be prompted.

Step 3: Choose the required upgrade firmware package and the upgrade process will start automatically. Wait for the upgrade to complete, APP will disconnect with Beacon after upgrade completed successfully.

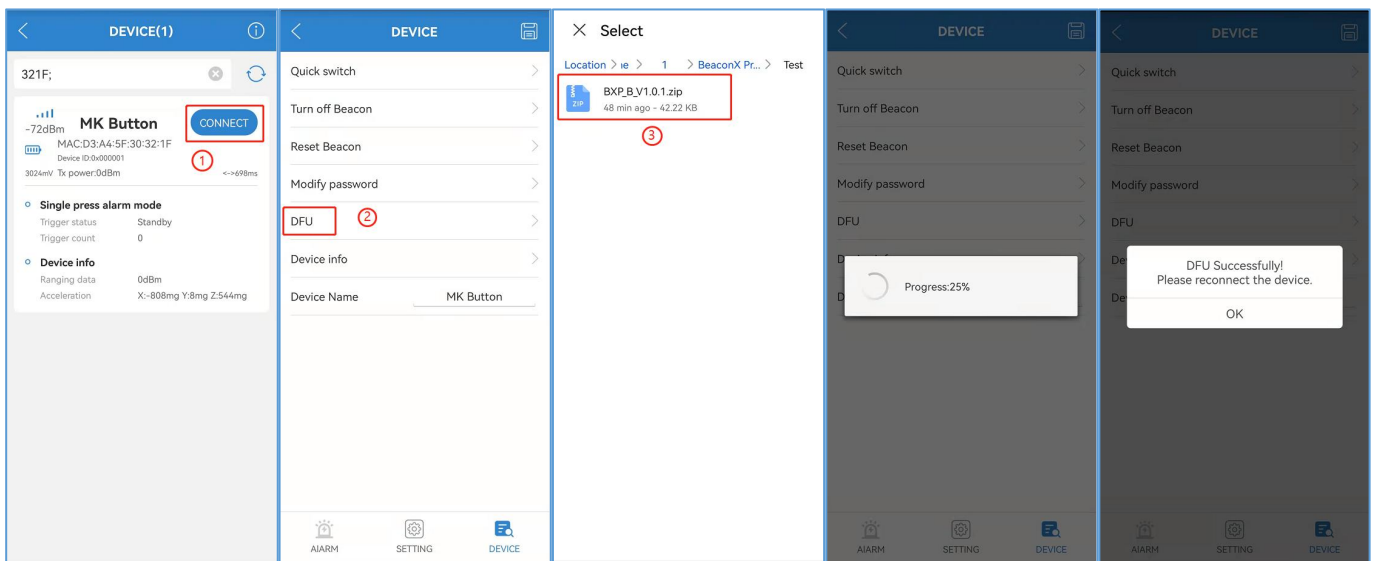


Figure 20: Firmware upgrade workflow of MOKO TAG application (Android)

3.13.2 iOS – MOKO APP DFU instructions

Step 1: Install iTunes in PC and then import firmware upgrade package into “**MOKO TAG**” APP, you can refer to below picture demonstrations – **Figure 35: Load upgrade package to nRF connect APP via iTunes.**

Step 2: Connect with device and go to “**SETTING**” interface, press “**DFU**” option and then firmware upgrade package will be shown automatically in DFU path.

Step 3: Choose the required upgrade firmware package and the upgrade process will start automatically. Wait for the upgrade to complete, APP will disconnect with Beacon after upgrade completed successfully.

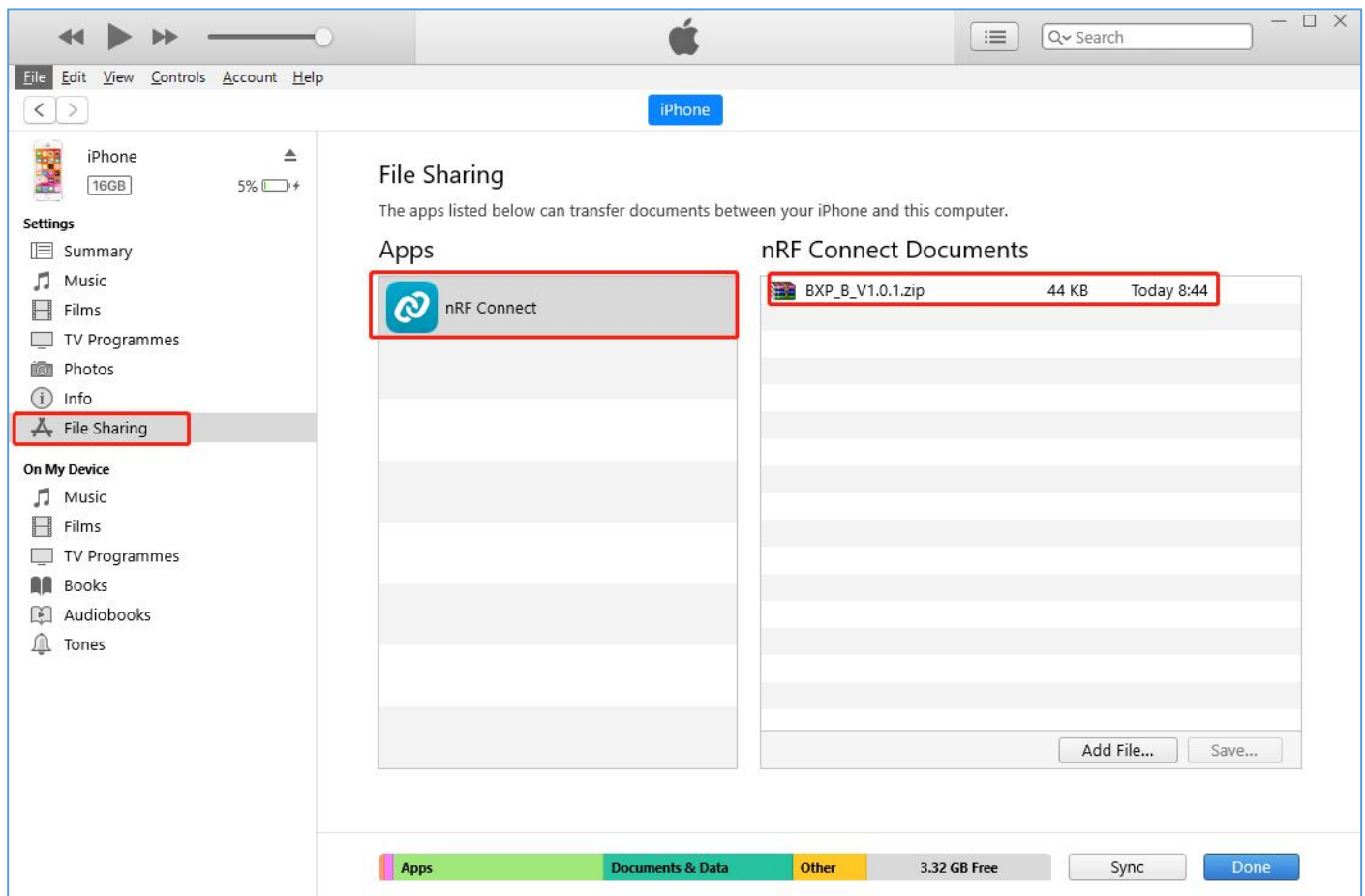


Figure 21: Load upgrade package to nRF connect APP via iTunes

3.14 Obtain log file from Android APP

Sometimes when testing with our APP, there may occurred some issues that cannot be reproduced in MOKO side, then we will need you to help obtain the log file from android APP and then provide for us to further analysis.

Currently only Android APP support log record, and the **LOG** option is in “**About**” interface, you can export the log file and then send to us by following below steps if rare issue happened.

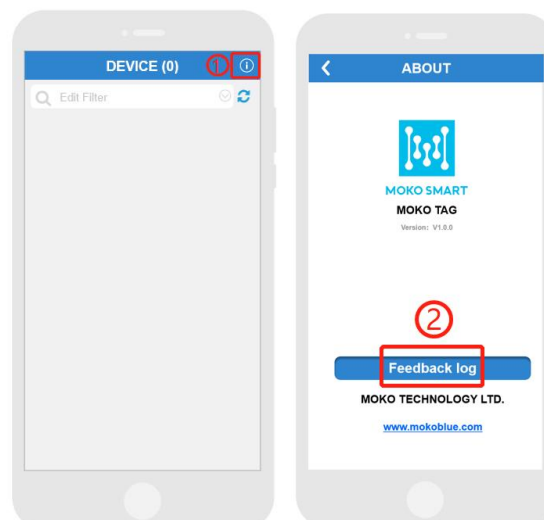


Figure 22: Obtain log file from Android APP

4. FAQ

General Questions

● What is RSSI?

The RSSI (Received Signal Strength Indication) is a measurement of the power present in a received radio signal. It is usually displayed in dBm (Decibel-milliwatts). Low power devices, like Bluetooth Low Energy devices, are using very weak power to transfer data, therefore the usual values of RSSI are between -100dBm (Very weak signal) to around -20 dBm (strong signal).

● Why the RSSI values are different on two Android devices.

By definition the RSSI value is a measurement of the power present in received radio signal. The signal strength depends on the antennas, device orientation, location, distance from the phone to target device, humidity etc.

● What's the difference between Advertising data and Scan Response data?

In Bluetooth 4+ device, if not connected, continuously advertises with the Advertising data. When a scanner receives it, it may send the Scan request packet to obtain more advertising information. Then, the advertising device sends the Scan response data. Both packets may have up to 31 bytes.

APP Questions

● Why the device name shows N/A?

Device name is contained in scan response packet in "Tag info" advertisement frame type, if "Tag info" slot is not set, then APP will not achieve the device name and will show N/A accordingly.

Appendix

A.1 Configuration range of overall parameters

Frame type	Parameters	Configuration range	Remark
iBeacon	UUID	16 bytes (Hex)	
	Major	0-65535 (Dec)	
	Minor	0-65535 (Dec)	
Eddystone - UID	Namespace ID	10 bytes (Hex)	
	Instance ID	6 bytes (Hex)	
Eddystone - URL	URL Link	1-17 characters (ASCII)	
Tag info	Device name	1-20 characters (ASCII)	
	Tag ID	1-6 bytes (Hex)	
General configurations	Adv interval	100ms – 10000ms	
	Adv duration	1s – 65535s	
	Standby duration	0s – 65535s	When standby duration set in 0, Adv duration will be ignored.
	Ranging data	-100dBm - 0dBm	
	Tx Power	-20/-16/-12/-8/-4/0/+3/+4/+6dBm	
	Connection password	1-16 characters (ASCII)	

Figure 23: Configuration range of overall parameters